

Titration report



Method

Method ID	CDC_GranTED-DYN-KHP-M002	Release state	Released
Name	Titr KHP with NaOH based on M002	Modified by	PredefinedUser
Modified on	3/9/2026 7:25:14 AM		

Summary

Samples

	Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.036399 mL	OK
		Concentration R2	0.10073 mol/L	OK
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.029497 mL	OK
		Concentration R2	0.10059 mol/L	OK
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL	
		Consumption R1	5.032976 mL	OK
		Concentration R2	0.10066 mol/L	OK

Statistics

R1

Mean	5.032957 mL	Minimum	5.029497 mL
Standard deviation	0.003451 mL	Rel. standard deviation	0.069 %
Total samples	3	Samples excluded	0

R2

Mean	0.10066 mol/L	Minimum	0.10059 mol/L
Standard deviation	0.00007 mol/L	Rel. standard deviation	0.070 %
Total samples	3	Samples excluded	0

Task

Task internal ID	T165502	Task name	DYN 0.02-0.2 mL VDISP=0 mL - M002 - KHP/NaOH
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Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.036399 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10073 mol/L	OK

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0.1 M TRIS - 5 mL in 60 mL H2O	Consumption	R1	5.029497 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10059 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Consumption	R1	5.032976 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10066 mol/L	OK

Raw Results

Sample scope: Scope 1/1, Sample 1/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	3/9/2026 7:25:47 AM
Overall state	OK
Sample start time	3/9/2026 7:25:47 AM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	3/9/2026 7:25:49 AM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	3/9/2026 7:25:50 AM
Overall state	OK
VENDDi	5 mL
QENDDi	0.50 mmol
tDi	00:07 min

Stir (Stir1)

Creation time	3/9/2026 7:26:07 AM
Overall state	OK

Titration (EQP) [1] (TitrationEQP1)

Creation time	3/9/2026 7:26:22 AM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	-102.4 mV
EHNv1	94.0 mV
Q1	0.503640 mmol
VEQ1	5.036399 mL
EST	159.0 mV
VEX	0.419601 mL
VEND	5.4560 mL
t	03:20 min
QEND	0.545600 mmol
QEX	0.041960 mmol

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EPD 159.1 mV
nEQ 1

Consumption (R1)

Creation time 3/9/2026 7:29:48 AM
Overall state OK
Result 5.036399 mL
Decimal places 6
Substance None
M 1 g/mol
z 1

Concentration (R2)

Creation time 3/9/2026 7:29:48 AM
Overall state OK
Result 0.10073 mol/L
Decimal places 5
Substance None
M 1 g/mol
z 1

Sample scope: Scope 1/1, Sample 2/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time 3/9/2026 7:34:46 AM
Overall state OK
Sample start time 3/9/2026 7:34:46 AM
Sample size 5 mL
Correction factor 1.0
Temperature 25.0 °C
Density 1.0 g/mL

Titration stand (Titrationstand1)

Creation time 3/9/2026 7:34:47 AM
Overall state OK

Dispense (normal) [1] (Dispensenormal1)

Creation time 3/9/2026 7:34:48 AM
Overall state OK
VENDDi 5 mL
QENDDi 0.50 mmol
tDi 00:08 min

Stir (Stir1)

Creation time 3/9/2026 7:35:05 AM
Overall state OK

Titration (EQP) [1] (TitrationEQP1)

Creation time 3/9/2026 7:35:20 AM
Overall state OK
Terminated at Number of EQPs

Titration report

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EEQ1	-91.8 mV
EHNv1	94.4 mV
Q1	0.502950 mmol
VEQ1	5.029497 mL
EST	158.7 mV
VEX	0.405503 mL
VEND	5.4350 mL
t	03:19 min
QEND	0.543500 mmol
QEX	0.040550 mmol
EPD	158.9 mV
nEQ	1

Consumption (R1)

Creation time	3/9/2026 7:38:44 AM
Overall state	OK
Result	5.029497 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

Creation time	3/9/2026 7:38:44 AM
Overall state	OK
Result	0.10059 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Sample scope: Scope 1/1, Sample 3/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	3/9/2026 7:39:10 AM
Overall state	OK
Sample start time	3/9/2026 7:39:10 AM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	3/9/2026 7:39:11 AM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	3/9/2026 7:39:11 AM
Overall state	OK
VENDDi	5 mL
QENDDi	0.50 mmol
tDi	00:07 min

Titration report

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Stir (Stir1)

Creation time 3/9/2026 7:39:29 AM
Overall state OK

Titration (EQP) [1] (TitrationEQP1)

Creation time 3/9/2026 7:39:44 AM
Overall state OK
Terminated at Number of EQPs
EEQ1 -91.5 mV
EHN1 94.7 mV
Q1 0.503298 mmol
VEQ1 5.032976 mL
EST 158.9 mV
VEX 0.365024 mL
VEND 5.3980 mL
t 03:23 min
QEND 0.539800 mmol
QEX 0.036502 mmol
EPD 159.0 mV
nEQ 1

Consumption (R1)

Creation time 3/9/2026 7:43:12 AM
Overall state OK
Result 5.032976 mL
Decimal places 6
Substance None
M 1 g/mol
z 1

Concentration (R2)

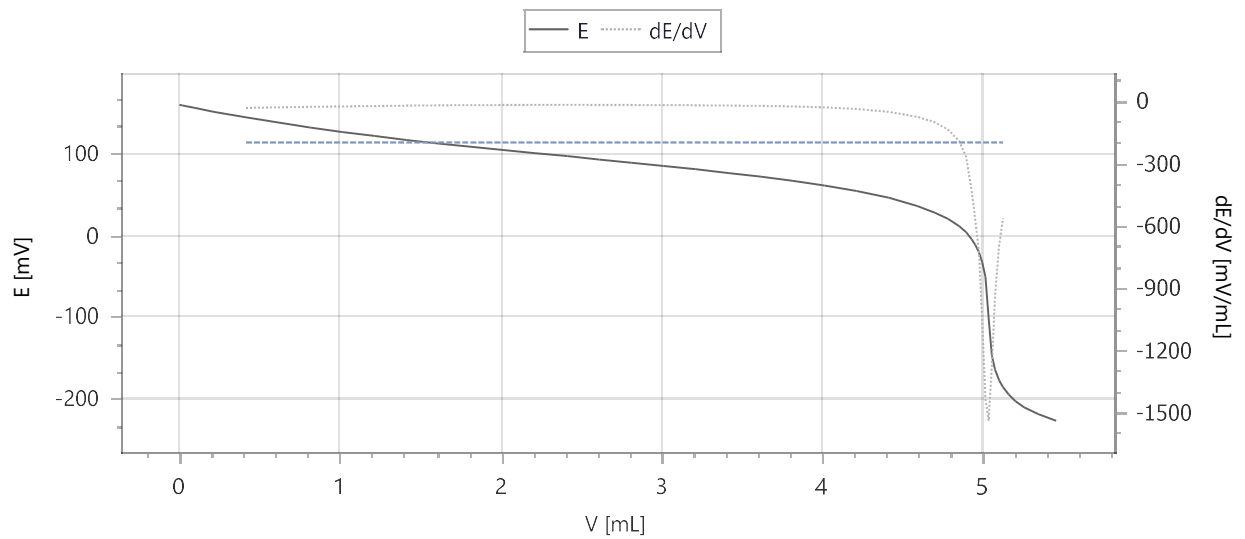
Creation time 3/9/2026 7:43:13 AM
Overall state OK
Result 0.10066 mol/L
Decimal places 5
Substance None
M 1 g/mol
z 1

Titration (EQP) [1] Scope 1/1, Sample 1/3

E - V & dE/dV - V Curve Scope 1/1, Sample 1/3

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Measured Values Scope 1/1, Sample 1/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	159.0		25.0			
33	0.0200	158.5	0.0200	25.0	-0.5		
36	0.0400	157.7	0.0200	25.0	-0.8		
40	0.0900	155.7	0.0500	25.0	-2.0		
43	0.2150	150.6	0.1250	25.0	-5.1		
46	0.4135	143.9	0.1985	25.0	-6.7	-33.71	
49	0.6135	137.6	0.2000	25.0	-6.3	-30.87	
52	0.8135	131.4	0.2000	25.0	-6.2	-28.43	
55	1.0135	125.9	0.2000	25.0	-5.5	-26.31	
58	1.2135	121.1	0.2000	25.0	-4.8	-24.48	
61	1.4135	116.2	0.2000	25.0	-4.9	-22.72	
64	1.6135	111.8	0.2000	25.0	-4.4	-21.07	
67	1.8135	107.7	0.2000	25.0	-4.1	-20.20	
70	2.0135	103.8	0.2000	25.0	-3.9	-19.59	
73	2.2135	99.9	0.2000	25.0	-3.9	-19.16	
76	2.4135	96.3	0.2000	25.0	-3.6	-19.11	
79	2.6135	92.0	0.2000	25.0	-4.3	-19.35	
82	2.8135	88.3	0.2000	25.0	-3.7	-19.61	
85	3.0135	84.4	0.2000	25.0	-3.9	-20.03	
88	3.2135	80.3	0.2000	25.0	-4.1	-20.94	
92	3.4135	75.8	0.2000	25.0	-4.5	-21.97	
94	3.6135	71.4	0.2000	25.0	-4.4	-23.78	
98	3.8135	66.2	0.2000	25.0	-5.2	-26.44	
101	4.0135	60.3	0.2000	25.0	-5.9	-30.93	
104	4.2135	53.9	0.2000	25.0	-6.4	-38.39	
107	4.4135	45.4	0.2000	25.0	-8.5	-52.47	
111	4.5920	35.5	0.1785	25.0	-9.9	-76.34	
114	4.7020	27.3	0.1100	25.0	-8.2	-102.84	
118	4.7820	19.9	0.0800	25.0	-7.4	-136.80	
122	4.8520	10.9	0.0700	25.0	-9.0	-193.02	
126	4.8960	3.3	0.0440	25.0	-7.6	-270.47	

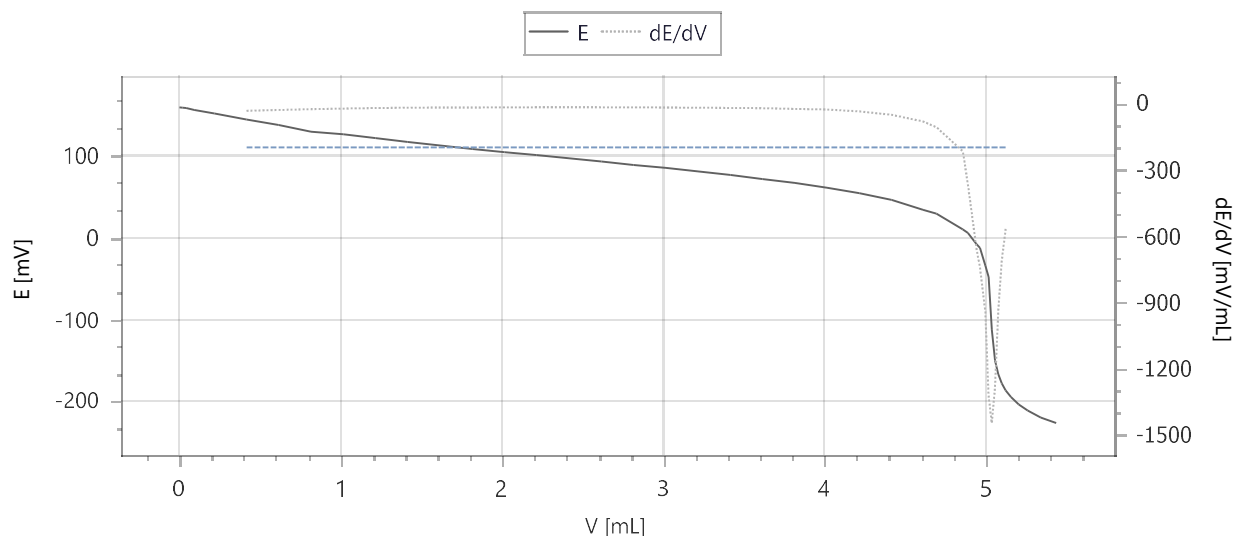
Titration report

LabX

129	4.9305	-5.3	0.0345	25.0	-8.6	-435.79	
132	4.9530	-12.0	0.0225	25.0	-6.7	-572.88	
135	4.9760	-21.4	0.0230	25.0	-9.4	-732.25	
138	4.9960	-33.4	0.0200	25.0	-12.0	-1058.95	
143	5.0160	-51.1	0.0200	25.0	-17.7	-1436.79	
152	5.0360	-101.5	0.0200	25.0	-50.4	-1541.01	
	5.036399	-102.4				-1542.49	EQP
160	5.0560	-147.4	0.0200	25.0	-45.9	-1276.62	
164	5.0760	-164.6	0.0200	25.0	-17.2	-936.48	
168	5.1010	-177.8	0.0250	25.0	-13.2	-696.38	
171	5.1255	-186.3	0.0245	25.0	-8.5	-565.29	
174	5.1605	-195.2	0.0350	25.0	-8.9		
177	5.2030	-203.4	0.0425	25.0	-8.2		
180	5.2580	-210.7	0.0550	25.0	-7.3		
183	5.3440	-219.1	0.0860	25.0	-8.4		
186	5.4560	-227.3	0.1120	25.0	-8.2		

Titration (EQP) [1] Scope 1/1, Sample 2/3

E - V & dE/dV - V Curve Scope 1/1, Sample 2/3



Measured Values Scope 1/1, Sample 2/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	158.7		25.0			
33	0.0200	158.0	0.0200	25.0	-0.7		
36	0.0400	157.8	0.0200	25.0	-0.2		
40	0.0900	155.5	0.0500	25.0	-2.3		
43	0.2150	151.1	0.1250	25.0	-4.4		
46	0.4150	143.7	0.2000	25.0	-7.4	-35.28	
49	0.6150	137.0	0.2000	25.0	-6.7	-31.37	
52	0.8150	129.0	0.2000	25.0	-8.0	-27.78	
55	1.0150	125.9	0.2000	25.0	-3.1	-25.02	
58	1.2150	120.9	0.2000	25.0	-5.0	-22.74	
61	1.4150	116.5	0.2000	25.0	-4.4	-21.07	

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LabX

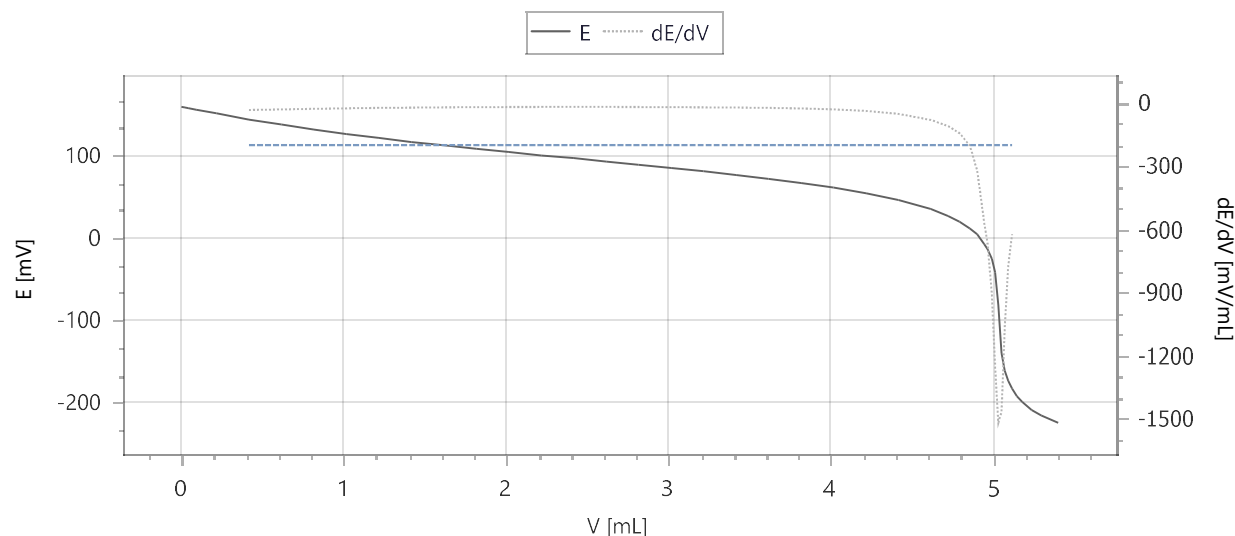
64	1.6150	111.9	0.2000	25.0	-4.6	-20.30	EQP
68	1.8150	107.8	0.2000	25.0	-4.1	-20.56	
71	2.0150	103.9	0.2000	25.0	-3.9	-19.46	
74	2.2150	100.4	0.2000	25.0	-3.5	-19.10	
77	2.4150	96.4	0.2000	25.0	-4.0	-18.98	
80	2.6150	92.4	0.2000	25.0	-4.0	-19.15	
83	2.8150	88.4	0.2000	25.0	-4.0	-19.79	
86	3.0150	84.8	0.2000	25.0	-3.6	-20.32	
89	3.2150	80.5	0.2000	25.0	-4.3	-21.01	
92	3.4150	76.2	0.2000	25.0	-4.3	-22.26	
95	3.6150	71.1	0.2000	25.0	-5.1	-23.42	
98	3.8150	66.5	0.2000	25.0	-4.6	-26.73	
101	4.0150	60.6	0.2000	25.0	-5.9	-29.28	
104	4.2150	54.0	0.2000	25.0	-6.6	-37.91	
107	4.4150	45.9	0.2000	25.0	-8.1	-53.21	
110	4.6100	33.8	0.1950	25.0	-12.1	-83.30	
113	4.6940	29.1	0.0840	25.0	-4.7	-109.80	
116	4.8560	9.9	0.1620	25.0	-19.2	-217.83	
120	4.8875	5.7	0.0315	25.0	-4.2	-364.26	
123	4.9420	-8.1	0.0545	25.0	-13.8	-662.73	
126	4.9620	-13.0	0.0200	25.0	-4.9	-739.28	
130	4.9960	-34.5	0.0340	25.0	-21.5	-943.74	
134	5.0160	-48.5	0.0200	25.0	-14.0	-1321.20	
	5.029497	-91.8				-1455.39	
147	5.0360	-112.7	0.0200	25.0	-64.2	-1446.07	
156	5.0560	-150.0	0.0200	25.0	-37.3	-1286.36	
161	5.0760	-166.6	0.0200	25.0	-16.6	-941.26	
165	5.0980	-178.0	0.0220	25.0	-11.4	-711.68	
168	5.1225	-186.6	0.0245	25.0	-8.6	-565.80	
172	5.1560	-194.8	0.0335	25.0	-8.2		
175	5.2030	-203.5	0.0470	25.0	-8.7		
178	5.2595	-211.2	0.0565	25.0	-7.7		
181	5.3395	-219.8	0.0800	25.0	-8.6		
184	5.4350	-226.4	0.0955	25.0	-6.6		

Titration (EQP) [1] Scope 1/1, Sample 3/3

E - V & dE/dV - V Curve Scope 1/1, Sample 3/3

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Measured Values Scope 1/1, Sample 3/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	158.9		25.0			
33	0.0200	158.2	0.0200	25.0	-0.7		
36	0.0400	157.5	0.0200	25.0	-0.7		
40	0.0900	155.7	0.0500	25.0	-1.8		
43	0.2150	151.1	0.1250	25.0	-4.6		
46	0.4150	143.6	0.2000	25.0	-7.5	-33.52	
49	0.6150	137.5	0.2000	25.0	-6.1	-30.89	
52	0.8150	131.4	0.2000	25.0	-6.1	-28.03	
55	1.0150	126.1	0.2000	25.0	-5.3	-25.65	
58	1.2150	121.3	0.2000	25.0	-4.8	-23.60	
61	1.4150	116.4	0.2000	25.0	-4.9	-22.39	
64	1.6150	112.5	0.2000	25.0	-3.9	-21.03	
67	1.8150	108.1	0.2000	25.0	-4.4	-20.19	
70	2.0150	104.2	0.2000	25.0	-3.9	-19.51	
73	2.2150	100.1	0.2000	25.0	-4.1	-19.15	
76	2.4150	96.7	0.2000	25.0	-3.4	-19.06	
79	2.6150	92.7	0.2000	25.0	-4.0	-19.03	
82	2.8150	88.8	0.2000	25.0	-3.9	-19.52	
85	3.0150	84.8	0.2000	25.0	-4.0	-20.30	
88	3.2150	80.8	0.2000	25.0	-4.0	-20.99	
91	3.4150	76.3	0.2000	25.0	-4.5	-22.00	
94	3.6150	71.7	0.2000	25.0	-4.6	-23.60	
98	3.8150	66.6	0.2000	25.0	-5.1	-26.20	
101	4.0150	61.3	0.2000	25.0	-5.3	-30.59	
104	4.2150	54.1	0.2000	25.0	-7.2	-37.89	
107	4.4150	46.1	0.2000	25.0	-8.0	-51.92	
110	4.6135	35.2	0.1985	25.0	-10.9	-80.71	
113	4.7205	26.8	0.1070	25.0	-8.4	-109.40	
116	4.7915	19.7	0.0710	25.0	-7.1	-144.07	
119	4.8540	11.4	0.0625	25.0	-8.3	-202.18	
122	4.8995	4.3	0.0455	25.0	-7.1	-320.33	

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125	4.9435	-7.7	0.0440	25.0	-12.0	-571.65	EQP
128	4.9635	-13.5	0.0200	25.0	-5.8	-668.69	
132	4.9895	-25.6	0.0260	25.0	-12.1	-902.73	
137	5.0095	-41.8	0.0200	25.0	-16.2	-1244.28	
149	5.0295	-81.4	0.0200	25.0	-39.6	-1517.63	
	5.032976	-91.5				-1517.64	
160	5.0495	-139.7	0.0200	25.0	-58.3	-1466.02	
166	5.0695	-160.8	0.0200	25.0	-21.1	-1063.59	
170	5.0905	-173.3	0.0210	25.0	-12.5	-774.06	
173	5.1145	-182.8	0.0240	25.0	-9.5	-623.23	
176	5.1445	-192.0	0.0300	25.0	-9.2		
180	5.1785	-199.1	0.0340	25.0	-7.1		
183	5.2345	-208.5	0.0560	25.0	-9.4		
186	5.2945	-215.3	0.0600	25.0	-6.8		
189	5.3980	-224.2	0.1035	25.0	-8.9		

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1

Sample ID 2

Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Task Parameter

Sample ID	Parameter name	Qualified name	Type	Value	Time
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	DYN 0.02-0.2 mL VDISP=0 mL - M002 - KHP/NaOH	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	09.03.2026 07:25:45

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0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	DYN 0.02-0.2 mL VDISP=0 mL - M002 - KHP/NaOH	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	09.03.2026 07:25:45
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	DYN 0.02-0.2 mL VDISP=0 mL - M002 - KHP/NaOH	09.03.2026 07:25:45

Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Responsible User

Time	Login name	Full name
09.03.2026 07:25:45	PredefinedUser	Administrator